

Week 6
Velocity Kinematics

ENN581 Robot Motion, Control and Planning

Dr. Krishna Manaswi Digumarti

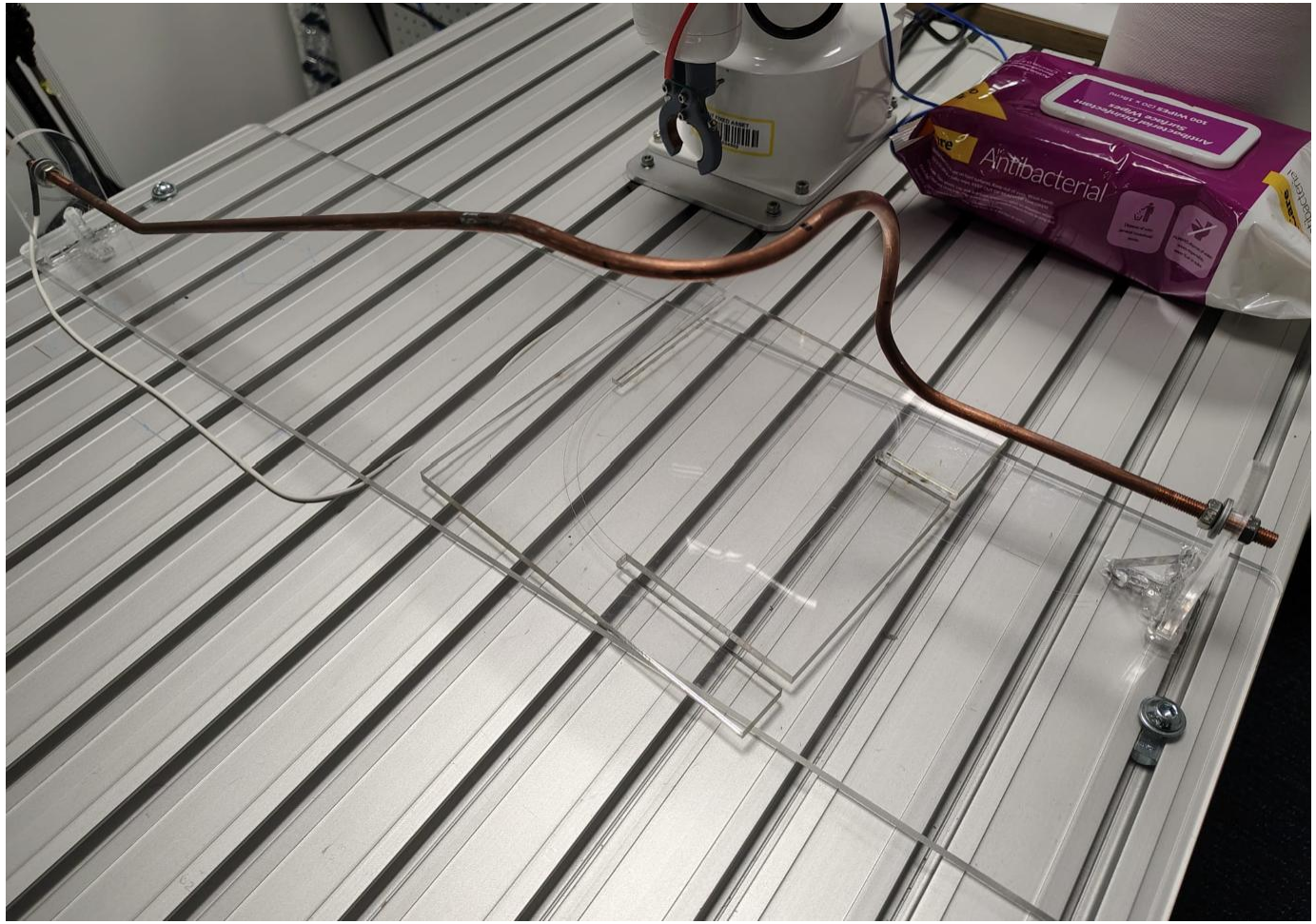
School of Electrical Engineering and Robotics

ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

How is the assessment going?



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Check submitted files (0/0)

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All required files submitted!
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A2.3 Q3.1 Test 1) Test that the generateLinearTrajectory method can be imported (0/0)

A2.3 Q3.1 Test 2) Test that the generateLinearTrajectory method returns appropriate shape of q (0/0)

```
Test Failed: No valid IK solution found within joint limits.
```

A2.3 Q3.1 Test 3) Test the generateLinearTrajectory method on randomly generated data (1/1)

A2.3 Q3.2 Test 1) Test that the generateCircularTrajectory method can be imported (0/0)

A2.3 Q3.2 Test 2) Test that the generateCircularTrajectory method returns appropriate shape of q (0/0)

```
Test Failed: No valid IK solution found within joint limits.
```

A2.3 Q3.2 Test 3) Test the generateCircularTrajectory method on randomly generated data (1/1)

A2.3 Q3.3 Test 1) Test that the jointAnglesAtTestPoints method can be imported (0/0)

A2.3 Q3.3 Test 2) Test that the jointAnglesAtTestPoints method returns correct shape of q (0/0)

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What did we learn so far?

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Today's lecture

- Velocities of manipulators – extend our study beyond static positioning problems
  - Linear
  - Angular
- Jacobian – a matrix that appears in the study of velocities and forces
- Singularities
- Manipulability

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The problem

- In a manipulator robot, we have direct control of actuators in joints
- How fast is the end effector moving, if each joint is moving at a certain velocity?

Aim:

To relate the linear and angular velocities of the end effector to joint velocities

How do we obtain $\begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{\omega} \end{bmatrix}$ if we are given $\dot{\mathbf{q}}$?

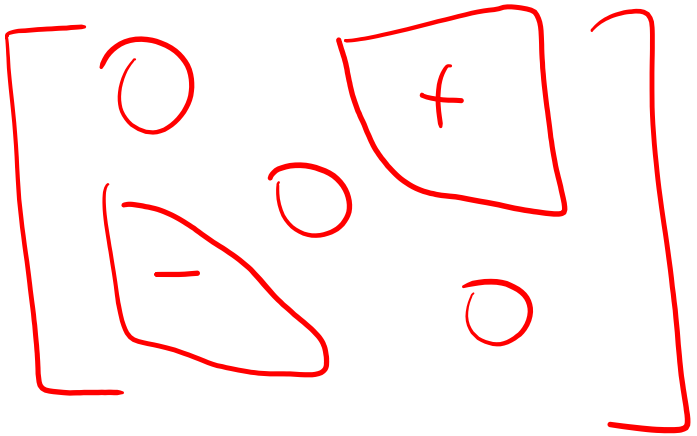
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Prerequisites

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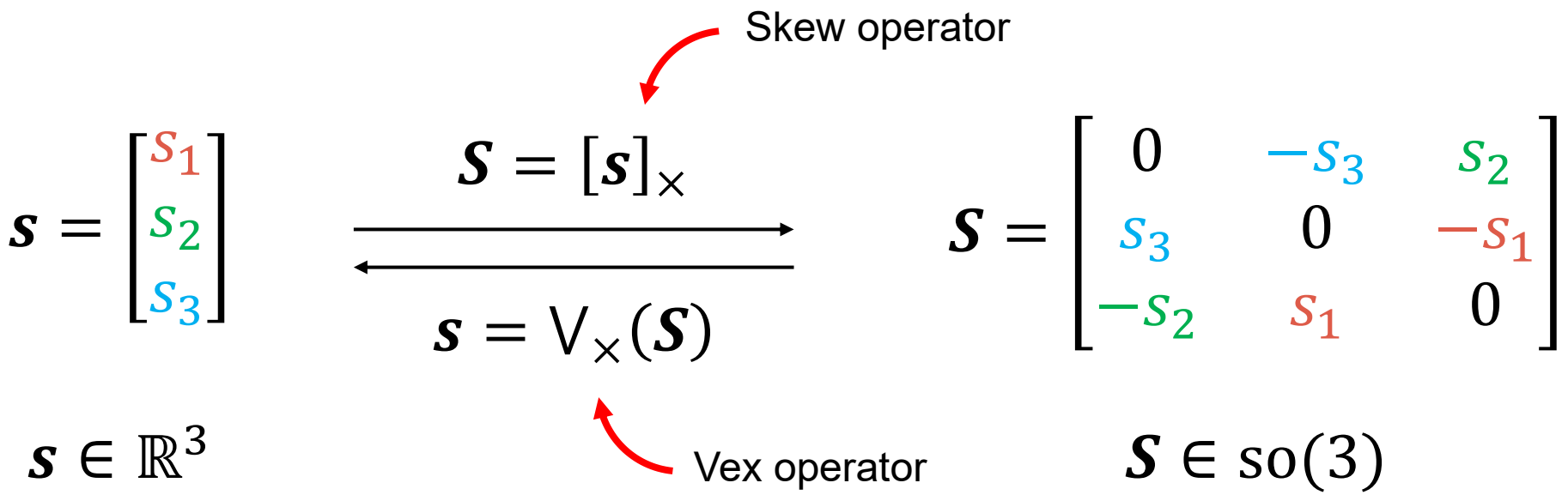
Skew symmetric matrix

$$A^T = -A$$



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Skew symmetric matrix



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Augmented skew symmetric matrix

4x4

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Augmented skew symmetric matrix

$$\hat{\boldsymbol{s}} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \end{bmatrix}$$

$$\hat{\boldsymbol{s}} \in \mathbb{R}^6$$

$$\hat{\boldsymbol{S}} = [\hat{\boldsymbol{s}}]_\times$$

$$\hat{\boldsymbol{s}} = \mathrm{V}_\times(\hat{\boldsymbol{S}})$$

$$\hat{\boldsymbol{S}} = \begin{bmatrix} 0 & -s_6 & s_5 & s_1 \\ s_6 & 0 & -s_4 & s_2 \\ -s_5 & s_4 & 0 & s_3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{\boldsymbol{S}} \in \mathrm{se}(3)$$

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Recall the elementary transform sequence

$$\dot{\tau} = \frac{d}{dt}(\tau)$$

$$^{0}\mathbf{T}_{\mathrm{E}} = \prod_{i=1}^{\mathrm{M}} \mathbf{E}_{i}(\eta_i)$$

Pose can be expressed as a product of M elementary transforms
This is what we have been using to describe the end effector pose

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What is the first derivative of pose wrt time?

$$\frac{d}{dt}(T)$$

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First derivative of pose wrt time

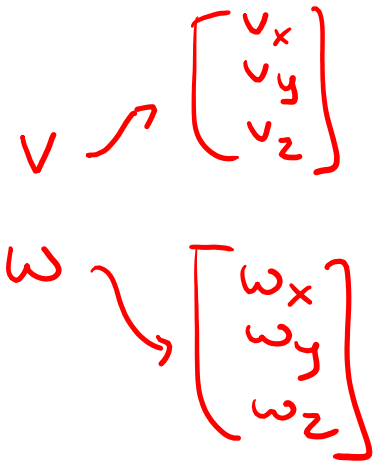
$$\dot{\mathbf{T}} = \frac{d\mathbf{T}}{dt} = \frac{\partial \mathbf{T}}{\partial q_1}\dot{q}_1 + \cdots + \frac{\partial \mathbf{T}}{\partial q_n}\dot{q}_n$$

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \qquad \dot{\mathbf{T}} = \begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{t}} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \qquad \frac{\partial \mathbf{T}}{\partial q_i} = \begin{bmatrix} \mathbf{J}_{\mathbf{R}_i} & \mathbf{J}_{\mathbf{t}_i} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$

$$\mathbf{T} \in \mathrm{SE}(3) \qquad \dot{\mathbf{T}} \in \mathbb{R}^{4\times 4} \qquad \frac{\partial \mathbf{T}}{\partial q_i} \in \mathbb{R}^{4\times 4} \qquad \mathbf{J}_{\mathbf{R}_i} \in \mathbb{R}^{3\times 3} \qquad \mathbf{J}_{\mathbf{t}_i} \in \mathbb{R}^{3\times 1}$$

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First derivative of pose wrt time



$$\dot{\mathbf{T}} = \frac{d\mathbf{T}}{dt} = \frac{\partial \mathbf{T}}{\partial q_1}\dot{q}_1 + \cdots + \frac{\partial \mathbf{T}}{\partial q_n}\dot{q}_n$$

$$\begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{t}} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_{\mathbf{R}_1} & \mathbf{J}_{\mathbf{t}_1} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \dot{q}_1 + \cdots + \begin{bmatrix} \mathbf{J}_{\mathbf{R}_n} & \mathbf{J}_{\mathbf{t}_n} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \dot{q}_n$$

$$\dot{\mathbf{R}} = \mathbf{J}_{\mathbf{R}_1}\dot{q}_1 + \cdots + \mathbf{J}_{\mathbf{R}_n}\dot{q}_n$$

$$\dot{\mathbf{t}} = \mathbf{J}_{\mathbf{t}_1}\dot{q}_1 + \cdots + \mathbf{J}_{\mathbf{t}_n}\dot{q}_n$$

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Consider the translation part

$$\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \dot{\mathbf{t}} = \mathbf{J}_{\mathbf{t}_1}\dot{q}_1 + \cdots + \mathbf{J}_{\mathbf{t}_n}\dot{q}_n$$

$$\dot{\mathbf{t}} = \begin{bmatrix} \mathbf{J}_{\mathbf{t}_1} & \cdots & \mathbf{J}_{\mathbf{t}_n} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

$$\boldsymbol{v} = \dot{\mathbf{t}} = \mathbf{J}_v(\mathbf{q})\dot{\mathbf{q}}$$

5 rad/s

10 °/s

100 mm/s

Translation part of the manipulator Jacobian

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Rotation part is a little more complex

$$\dot{\mathbf{R}} = \mathbf{J}_{\mathbf{R}_1}\dot{q}_1 + \cdots + \mathbf{J}_{\mathbf{R}_n}\dot{q}_n$$

$$[\boldsymbol{\omega}]_\times \mathbf{R} = \left(\mathbf{J}_{\mathbf{R}_1}\dot{q}_1 + \cdots + \mathbf{J}_{\mathbf{R}_n}\dot{q}_n\right)\mathbf{R}^\intercal$$

Post-multiply each term with $\mathbf{R}^\intercal$

$$[\boldsymbol{\omega}]_\times = (\mathbf{J}_{\mathbf{R}_1}\mathbf{R}^\intercal)\dot{q}_1 + \cdots + (\mathbf{J}_{\mathbf{R}_n}\mathbf{R}^\intercal)\dot{q}_n$$

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Rotation part is a little more complex

$$[\boldsymbol{\omega}]_\times = (\mathbf{J}_{\mathbf{R}_1}\mathbf{R}^\intercal)\dot{q}_1 + \cdots + (\mathbf{J}_{\mathbf{R}_n}\mathbf{R}^\intercal)\dot{q}_n$$

$$\boldsymbol{\omega} = \mathrm{V}_\times(\mathbf{J}_{\mathbf{R}_1}\mathbf{R}^\intercal)\dot{q}_1 + \cdots + \mathrm{V}_\times(\mathbf{J}_{\mathbf{R}_n}\mathbf{R}^\intercal)\dot{q}_n$$

$$\boldsymbol{\omega} = \begin{bmatrix} \mathrm{V}_\times(\mathbf{J}_{\mathbf{R}_1}\mathbf{R}^\intercal) & \cdots & \mathrm{V}_\times(\mathbf{J}_{\mathbf{R}_n}\mathbf{R}^\intercal) \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

$$\boldsymbol{\omega} = \mathbf{J}_\omega(\mathbf{q})\dot{\mathbf{q}}$$

3x1

Rotation part of the manipulator Jacobian

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The manipulator Jacobian

$$ {}^{0}\nu = \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_v(\mathbf{q}) \\ \mathbf{J}_\omega(\mathbf{q}) \end{bmatrix} \dot{\mathbf{q}} $$

$$ {}^{0}\nu = {}^{0}\mathbf{J}(\mathbf{q})\dot{\mathbf{q}} $$

$\dot{q} = \begin{Bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_{10} \end{Bmatrix}$

J 6x10

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The Manipulator Jacobian

Relates joint velocities to the cartesian velocities of the end effector.

$$^0\boldsymbol{\nu} = {}^0\mathbf{J}(\mathbf{q})\dot{\boldsymbol{q}}$$

6×1 $6 \times n$ $n \times 1$

Spatial velocity or twist — Jacobian — Vector of joint velocities

Pay attention to the frame → $^0\boldsymbol{\nu} = \begin{bmatrix} {}^0\boldsymbol{v} \\ {}^0\boldsymbol{\omega} \end{bmatrix}$

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What do the columns of a Jacobian mean?

Consider a 6 DoF robot manipulator in 3D space

$$\begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} j_{11} & j_{12} & j_{13} & j_{14} & j_{15} & j_{16} \\ j_{21} & j_{22} & j_{23} & j_{24} & j_{25} & j_{26} \\ j_{31} & j_{32} & j_{33} & j_{34} & j_{35} & j_{36} \\ j_{41} & j_{42} & j_{43} & j_{44} & j_{45} & j_{46} \\ j_{51} & j_{52} & j_{53} & j_{54} & j_{55} & j_{56} \\ j_{61} & j_{62} & j_{63} & j_{64} & j_{65} & j_{76} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{bmatrix}$$

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What do the columns of a Jacobian mean?

Consider a 6 DoF robot manipulator in 3D space

$$\begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} j_{11} & j_{12} & j_{13} & j_{14} & j_{15} & j_{16} \\ j_{21} & j_{22} & j_{23} & j_{24} & j_{25} & j_{26} \\ j_{31} & j_{32} & j_{33} & j_{34} & j_{35} & j_{36} \\ j_{41} & j_{42} & j_{43} & j_{44} & j_{45} & j_{46} \\ j_{51} & j_{52} & j_{53} & j_{54} & j_{55} & j_{56} \\ j_{61} & j_{62} & j_{63} & j_{64} & j_{65} & j_{76} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{bmatrix}$$

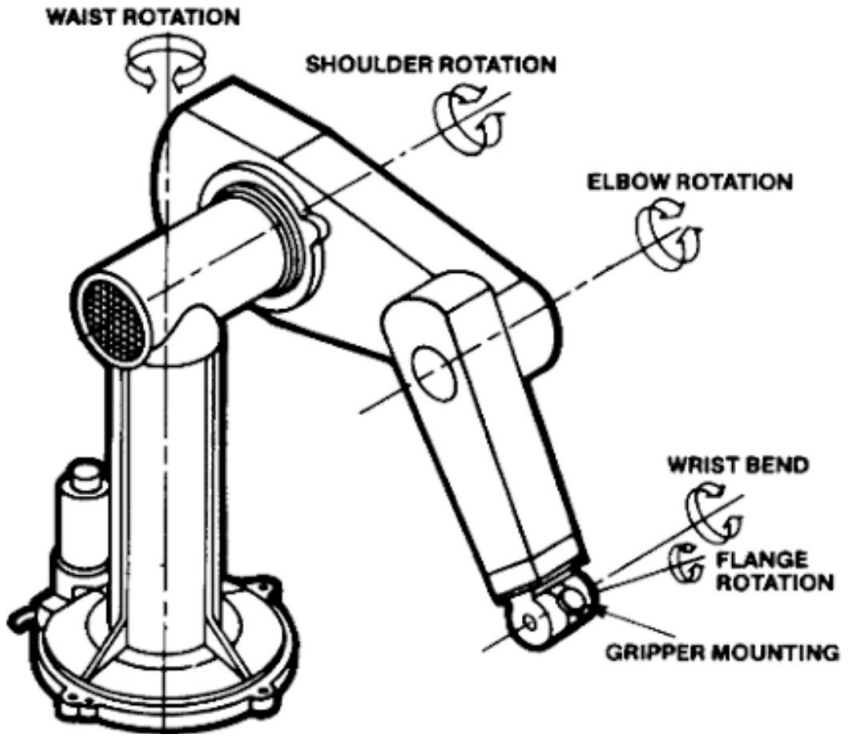
$\times \dot{q}_1 \quad \times \dot{q}_2$

What influence does the velocity of the first joint have on the spatial velocity of the manipulator

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Example

$$V = \begin{bmatrix} 0.13 & 0. & 0.306 & 0. & 0. & 0. \\ 0.664 & 0. & 0. & -0.038 & 0. & 0. \\ -0. & 0.664 & 0.359 & 0. & 0.053 & 0. \\ 0. & -0. & -0. & 0.707 & -0. & 1. \\ 0. & -1. & -1. & -0. & -1. & -0. \\ 1. & 0. & 0. & -0.707 & 0. & -0. \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_6 \end{bmatrix}$$



PUMA 560

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How do we compute $\mathbf{J}_v(\mathbf{q})$ and $\mathbf{J}_\omega(\mathbf{q})$?

$\dot{t} = J_t(q)\,\dot{q}$

$\dot{R} = J_R(q)\,\dot{q}$

$$\dot{\mathbf{T}} = \frac{d\mathbf{T}}{dt} = \frac{\partial \mathbf{T}}{\partial q_1}\dot{q}_1 + \cdots + \frac{\partial \mathbf{T}}{\partial q_n}\dot{q}_n$$

$$\mathbf{J}_v(\mathbf{q}) = \begin{bmatrix} \mathbf{J}_{t_1} & \cdots & \mathbf{J}_{t_n} \end{bmatrix}$$

$$\mathbf{J}_\omega(\mathbf{q}) = \begin{bmatrix} V_\times(\mathbf{J}_{R_1}\mathbf{R}^T) & \cdots & V_\times(\mathbf{J}_{R_n}\mathbf{R}^T) \end{bmatrix}$$

$$\begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{t}} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_{R_1} & \mathbf{J}_{t_1} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}\dot{q}_1 + \cdots + \begin{bmatrix} \mathbf{J}_{R_n} & \mathbf{J}_{t_n} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}\dot{q}_n$$

$$\dot{\mathbf{R}} = \mathbf{J}_{R_1}\dot{q}_1 + \cdots + \mathbf{J}_{R_n}\dot{q}_n$$

$$\dot{\mathbf{t}} = \mathbf{J}_{t_1}\dot{q}_1 + \cdots + \mathbf{J}_{t_n}\dot{q}_n$$

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How do we compute $\mathbf{J}_v(\mathbf{q})$ and $\mathbf{J}_\omega(\mathbf{q})$?

$$\mathbf{J}_v(\mathbf{q})$$

$$\mathbf{J}_{\mathbf{t}_j}(\mathbf{q}) \qquad j = 1 \cdots n$$

$$\tau\left(\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_j}\right)$$

Translation part of

$$\mathbf{J}_\omega(\mathbf{q})$$

$$\mathrm{V}_\times(\mathbf{J}_{\mathbf{R}_j}(\mathbf{q})\mathbf{R}(\mathbf{q})^\intercal) \qquad j = 1 \cdots n$$

$$\mathrm{V}_\times\left(\rho\left(\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_j}\right)\rho(\mathbf{T}(\mathbf{q}))^\intercal\right)$$

Rotation part of

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We need to compute $\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_j}$

$T = E_1 E_2 \ldots E$

Let us first learn how to compute the derivative of a single elementary transform wrt the joint coordinate

$$\frac{\partial \mathbf{E}_{\mu(j)}(q_j)}{\partial q_j}$$

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Derivative of a pure translation matrix wrt joint coordinate:

$$t_x(d) = \begin{bmatrix} 1 & 0 & 0 & d \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

d: variable $\sim q_1$

$$f(x) = x$$

$$\frac{d}{dx}(f(x)) = 1$$

$$\frac{d}{dq_1} t_x(q_1) = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\frac{dT}{dt} = \frac{\partial T}{\partial q_1} \dot{q}_1 + \dots$$

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Derivative of a pure translation matrix wrt joint coordinate:

$$\frac{\partial \mathbf{T}_{\mathbf{t}_x}(d)}{\partial d} =$$

$$\frac{\partial \mathbf{T}_{\mathbf{t}_y}(d)}{\partial d} =$$

$$\frac{\partial \mathbf{T}_{\mathbf{t}_z}(d)}{\partial d} =$$

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Derivative of a pure rotation matrix wrt joint coordinate:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\frac{\partial}{\partial\theta} R_x(\theta) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -s\theta & -c\theta & 0 \\ 0 & c\theta & -s\theta & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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Derivative of a pure rotation matrix wrt joint coordinate:

$$\frac{\partial \mathbf{T}_{\mathbf{R}_x}(\theta)}{\partial \theta} =$$

$$\frac{\partial \mathbf{T}_{\mathbf{R}_y}(\theta)}{\partial \theta} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{T}_{\mathbf{R}_y}(\theta) = \left[\widehat{\mathbf{R}_y}\right]_\times \mathbf{T}_{\mathbf{R}_y}(\theta)$$

$$\frac{\partial \mathbf{T}_{\mathbf{R}_z}(\theta)}{\partial \theta} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mathbf{T}_{\mathbf{R}_z}(\theta) = \left[\widehat{\mathbf{R}_z}\right]_\times \mathbf{T}_{\mathbf{R}_z}(\theta)$$

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Now we can compute $\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_j}$

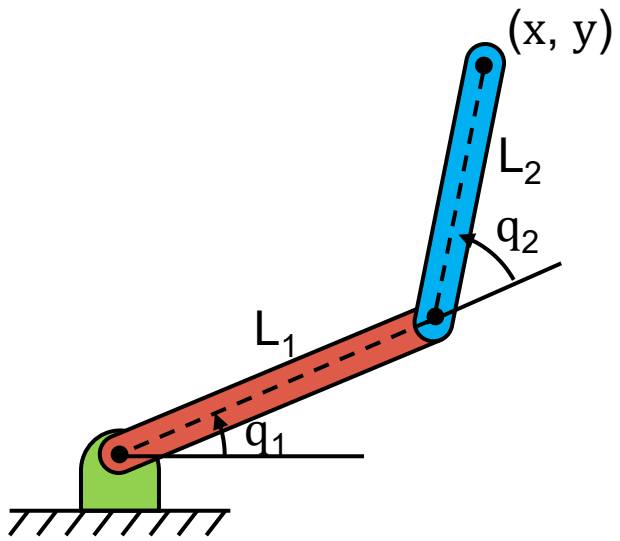
$$\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_j} = \frac{\partial}{\partial q_j}\left(\mathbf{E}_1(\eta_1)\cdots\mathbf{E}_n(\eta_n)\right)$$

$$= \prod_{i=1}^{\mu(j)-1} \mathbf{E}_i(\eta_i)\,\frac{\partial \mathbf{E}_{\mu(j)}(q_j)}{\partial q_j}\,\prod_{i=\mu(j)+1}^{M} \mathbf{E}_i(\eta_i)$$

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Example: 2 link planar RR manipulator

Express (x, y) in terms of joint coordinates using FK. Then compute the time derivative.



$$x = L_1 \cos q_1 + L_2 \cos(q_1 + q_2)$$

$$y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

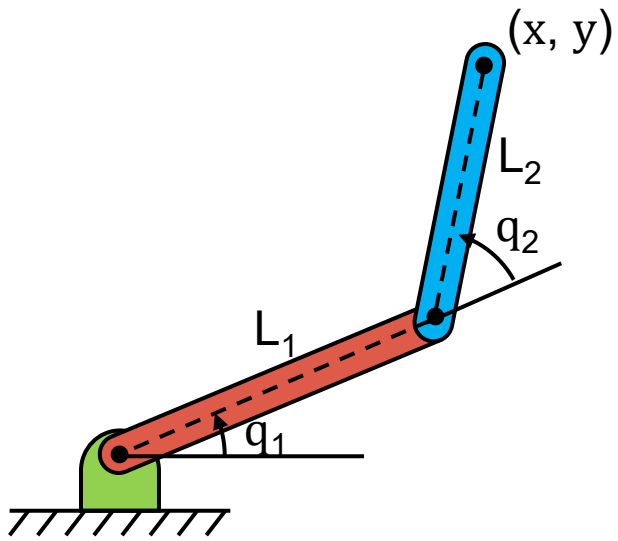
$$\dot{x} = f(q_1, \dot{q}_1) = \frac{d}{dt}(x) =$$

$$\dot{y} = g(q_1, \dot{q}_1)$$

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Example: 2 link planar RR manipulator

Express (x, y) in terms of joint coordinates using FK. Then compute the time derivative.



$$x = L_1 \cos(q_1) + L_2 \cos(q_1 + q_2)$$

$$y = L_1 \sin(q_1) + L_2 \sin(q_1 + q_2)$$

$$\dot{x} = -L_1 \sin(q_1)\dot{q_1} - L_2 \sin(q_1 + q_2)(\dot{q_1} + \dot{q_2})$$

$$\dot{y} = L_1 \cos(q_1)\dot{q_1} + L_2 \cos(q_1 + q_2)(\dot{q_1} + \dot{q_2})$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} \dot{q_1} \\ \dot{q_2} \end{bmatrix}$$

This is easy to do for this simple manipulator

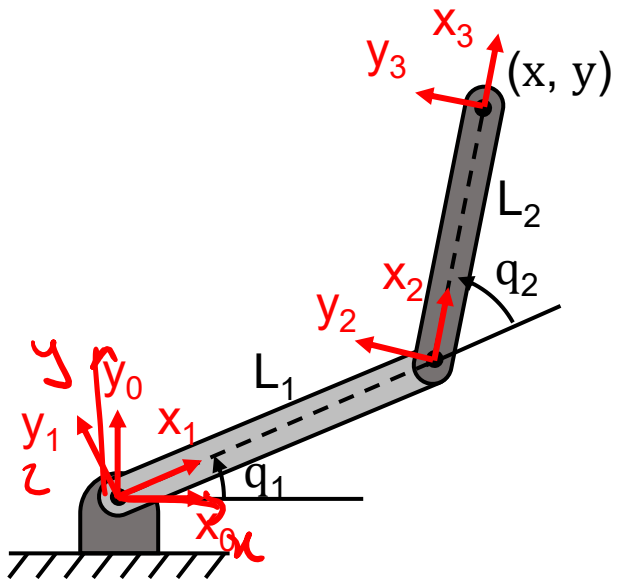
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Example: 2 link planar RR manipulator

Alternative approach: Let us apply the method we just learnt

$T = R_z(q_1)\, t_x(L_1)\, R_z(q_2)\, t_x(L_2)$

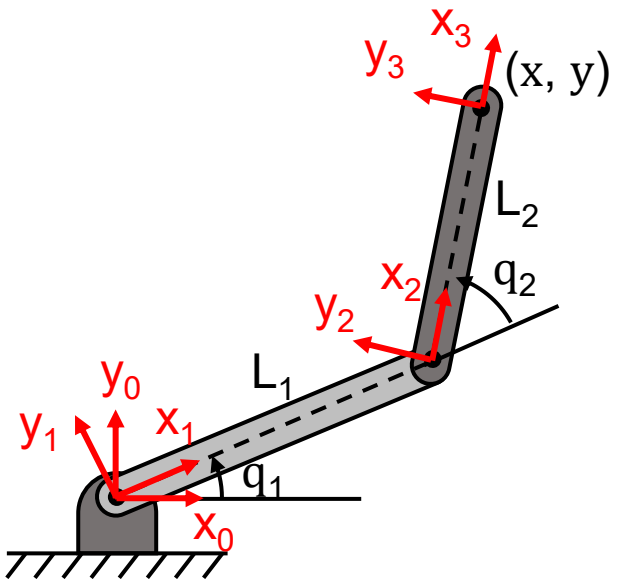
Step 1: Write down the ETS



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Example: 2 link planar RR manipulator

Alternative approach: Let us apply the method we just learnt



Step 1: Write down the ETS

$$
{}^{0}\mathbf{T}_3 = \underbrace{\mathbf{T}_{\mathbf{R}_z}(q_1)}_{\mathbf{E}_1}\underbrace{\mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_1)}_{\mathbf{E}_2}\underbrace{\mathbf{T}_{\mathbf{R}_z}(q_2)}_{\mathbf{E}_3}\underbrace{\mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_2)}_{\mathbf{E}_4}
$$

$$
\underbrace{\mathbf{T}_{\mathbf{R}_z}(q_1)}_{{}^{0}\mathbf{T}_1}\underbrace{\mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_1)\mathbf{T}_{\mathbf{R}_z}(q_2)}_{{}^{1}\mathbf{T}_2}\underbrace{\mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_2)}_{{}^{2}\mathbf{T}_3}
$$

$E_1 \rightarrow q_1$ $\quad$ $E_3 \rightarrow q_2$

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Step 2: Compute $\frac{\partial \mathbf{E}_{\mu(j)}(q_j)}{\partial q_j}$

For the first joint

$j = 1$

$\mu(1) = 1$

$$\frac{\partial \mathbf{E}_1(q_1)}{\partial q_1} = \frac{\partial \mathbf{T}_{\mathbf{R}_z}(q_1)}{\partial q_1} = \left[\widehat{\mathbf{R}_z}\right]_\times \mathbf{T}_{\mathbf{R}_z}(q_1)$$

For the second joint

$j = 2$

$\mu(2) = 3$

Because q_2 appears in $\mathbf{E}_3$

$$\frac{\partial \mathbf{E}_3(q_2)}{\partial q_2} = \frac{\partial \mathbf{T}_{\mathbf{R}_z}(q_2)}{\partial q_2} = \left[\widehat{\mathbf{R}_z}\right]_\times \mathbf{T}_{\mathbf{R}_z}(q_2)$$

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Step 3: Compute $\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_j}$

$$= \prod_{i=1}^{\mu(j)-1} \mathbf{E}_i(\eta_i) \frac{\partial \mathbf{E}_{\mu(j)}(q_j)}{\partial q_j} \prod_{i=\mu(j)+1}^{M} \mathbf{E}_i(\eta_i)$$

$$\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_1} = \frac{\partial \mathbf{E}_1(q_1)}{\partial q_1} \mathbf{E}_2 \mathbf{E}_3 \mathbf{E}_4 = \left[\widehat{\mathbf{R}_z}\right]_\times \mathbf{T}_{\mathbf{R}_z}(q_1) \mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_1) \mathbf{T}_{\mathbf{R}_z}(q_2) \mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_2)$$

$$\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_2} = \mathbf{E}_1 \mathbf{E}_2 \frac{\partial \mathbf{E}_3(q_2)}{\partial q_2} \mathbf{E}_4 = \mathbf{T}_{\mathbf{R}_z}(q_1) \mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_1) \left[\widehat{\mathbf{R}_z}\right]_\times \mathbf{T}_{\mathbf{R}_z}(q_2) \mathbf{T}_{\mathbf{t}_x}(\mathrm{L}_2)$$

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Step 4: Compute $\mathbf{J}_v(\mathbf{q})$ and $\mathbf{J}_\omega(\mathbf{q})$

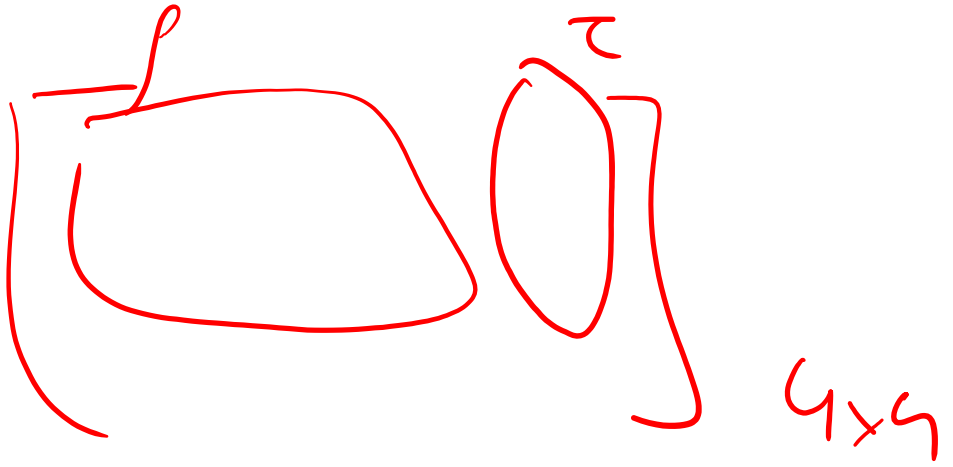
$$\mathbf{J}_v(\mathbf{q}) = [\mathbf{J}_{t_1}(\mathbf{q}) \quad \mathbf{J}_{t_2}(\mathbf{q})]$$

$$= \left[\tau\left(\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_1}\right) \quad \tau\left(\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_2}\right)\right]$$

$$\mathbf{J}_\omega(\mathbf{q}) = \left[\mathrm{V}_\times(\mathbf{J}_{\mathbf{R}_1}(\mathbf{q})\mathbf{R}(\mathbf{q})^\intercal) \quad \mathrm{V}_\times(\mathbf{J}_{\mathbf{R}_2}(\mathbf{q})\mathbf{R}(\mathbf{q})^\intercal)\right]$$

$$= \left[\mathrm{V}_\times\left(\rho\left(\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_1}\right)\rho(\mathbf{T}(\mathbf{q}))^\intercal\right) \quad \mathrm{V}_\times\left(\rho\left(\frac{\partial \mathbf{T}(\mathbf{q})}{\partial q_2}\right)\rho(\mathbf{T}(\mathbf{q}))^\intercal\right)\right]$$





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Changing the reference frame of the Jacobian

Say we are given a Jacobian in frame {B}, i.e. $\begin{bmatrix} {}^{B}v \\ {}^{B}\omega \end{bmatrix} = {}^{B}J(q)\dot{q}$

To express the Jacobian with respect to another frame {A}, the relation is

$$ {}^{A}J(q) = \begin{bmatrix} {}^{A}R_{B} & 0_{3\times3} \\ 0_{3\times3} & {}^{A}R_{B} \end{bmatrix} {}^{B}J(q) $$

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Inverting the Jacobian

Given $\dot{q}$

Find $\begin{bmatrix} v \\ \omega \end{bmatrix}$

- A common task in robotics is to be able to move the end effector at a desired velocity.
- In this scenario, we would like to know the appropriate joint velocities that move the end effector as desired.
- This is similar to the inverse kinematics problem for position.

$$\dot{q} = J^{-1}(q)\, \nu$$

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Is the Jacobian invertible?

- Is it a square matrix? If not, go for pseudo inverse. If yes,
- Is it non-singular? If not, inverse can't be computed. If yes, the inverse exists but,
- Is it invertible for all values of q? For what values of q is it not invertible?

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Analysis of Singularities

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Singularities

- The values of $\mathbf{q}$ where the Jacobian becomes singular i.e., its determinant is 0.

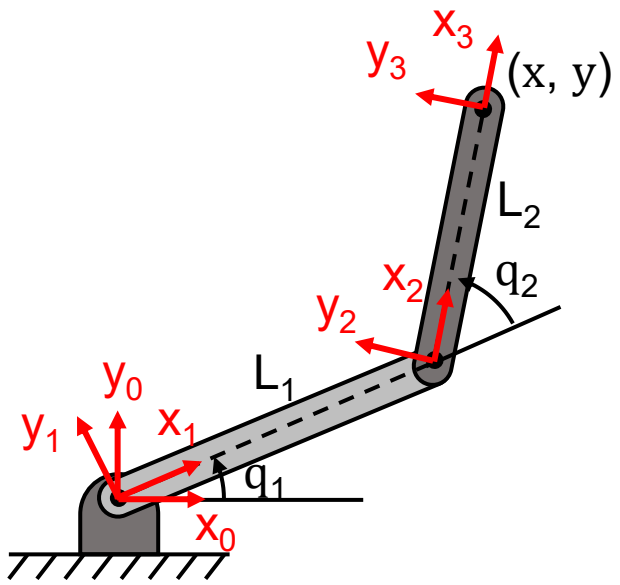
- Workspace-boundary singularities:
  - Occur at the boundaries of the workspace
  - Think of the manipulator being stretched out or folded in on itself
- Workspace interior singularities:
  - Occur away from the boundary of the workspace
  - Think of joints in a wrist lining up with each other

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Example: 2 link planar RR manipulator

det(Yellow) = $L_1 L_2 s q_2$

Determine the singularities of the manipulator



$$^3J(q) = \begin{bmatrix} L_1 s q_2 & 0 \\ L_1 c q_2 + L_2 & L_2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

Corresponds to v_z, ignore in 2D case
Corresponds to ω_x, ignore in 2D case
Corresponds to ω_y, ignore in 2D case

Compute the determinant

$$\begin{vmatrix} L_1 s q_2 & 0 \\ L_1 c q_2 + L_2 & L_2 \end{vmatrix} = L_1 L_2 s q_2$$

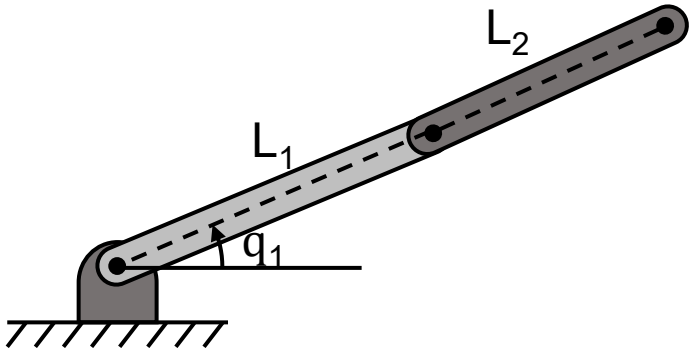
When is the determinant = 0?

At $q_2 = 0^\circ$ and at $q_2 = 180^\circ$

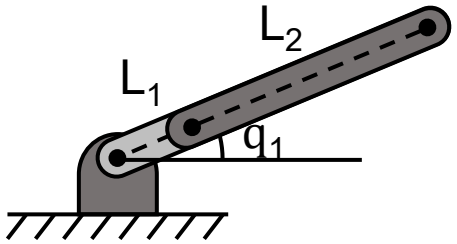
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What happens at these singularities?

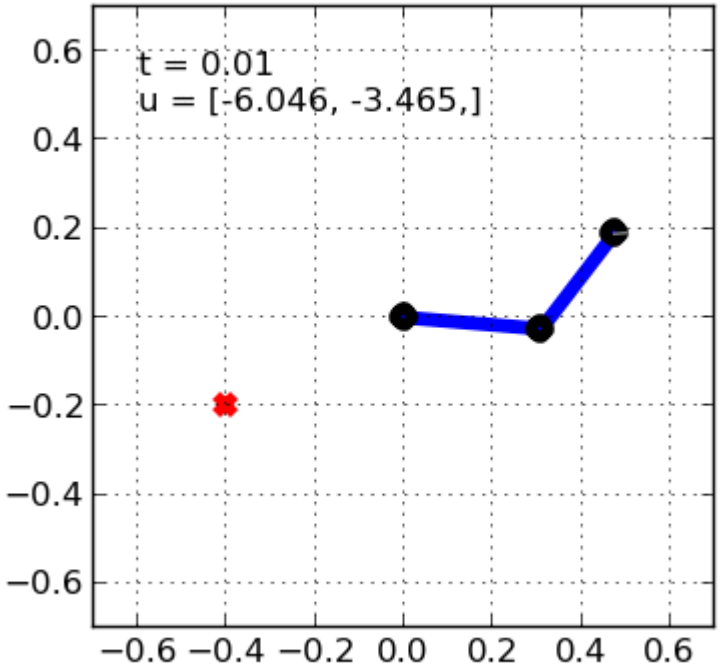
At $q_2 = 0^\circ$



At $q_2 = 180^\circ$



Demonstration of singularity



Credit: https://studywolf.wordpress.com/2013/10/10/robot-control-part-6-handling-singularities/

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Example

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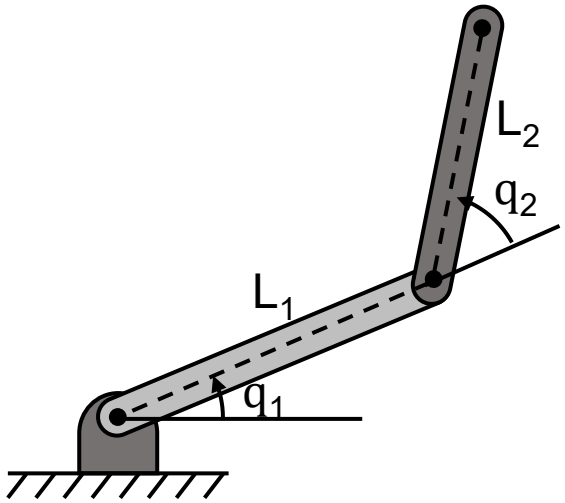
Manipulability

Given a joint configuration for the robot,
the capacity to change the position and orientation of the end effector.

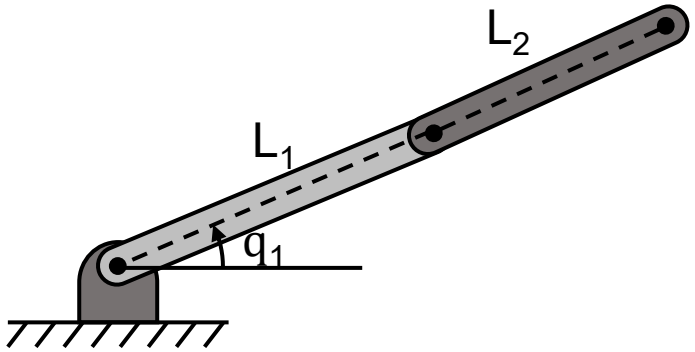
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We would like to know how close the robot is to a singularity

Far from a singularity



Close to a singularity

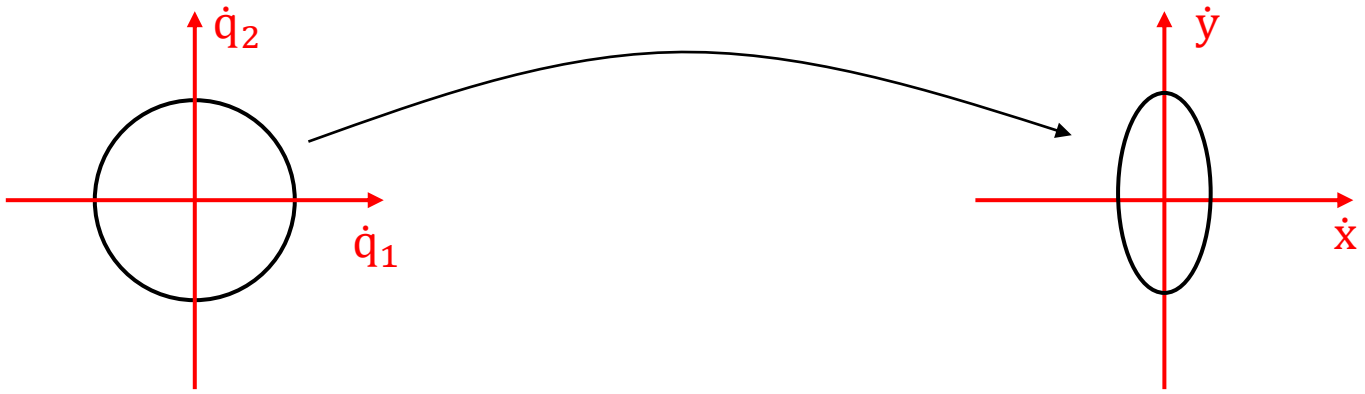


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Manipulability can be described as an ellipsoid

Consider a robot with n joints. Consider the set of all joint velocities with unit norm. $\dot{\mathbf{q}}\dot{\mathbf{q}}^{\mathsf{T}} = 1$

The set of all possible joint velocities can be thought of as a ball in n-dimensional space. These can be mapped to corresponding spatial velocities.



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An expression for the manipulability ellipsoid

We know that the joint velocities and the spatial velocities are related as

$$\boldsymbol{\nu} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$$

The manipulability ellipsoid is defined as the set of all velocities that satisfies

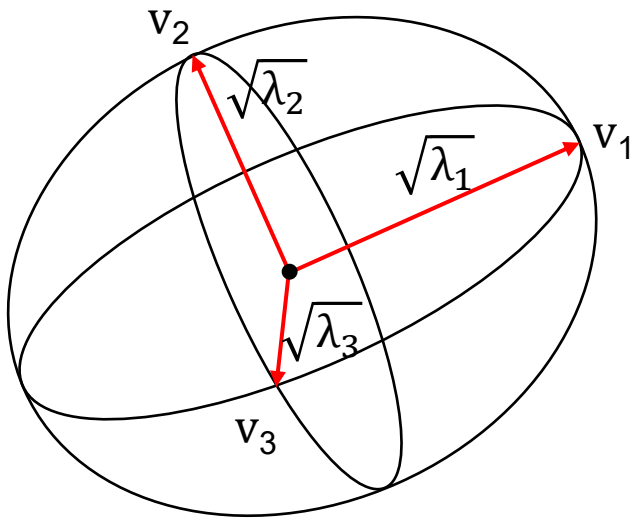
$$\boldsymbol{\nu}^{\mathsf{T}}\left(\mathbf{J}(\mathbf{q})\mathbf{J}^{\mathsf{T}}(\mathbf{q})\right)^{-1}\boldsymbol{\nu} \leq 1$$

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Determining the size of the ellipsoid

The ellipsoid is the region enclosed within $\boldsymbol{\nu}^{\mathsf{T}}\left(\mathbf{J}(\mathbf{q})\mathbf{J}^{\mathsf{T}}(\mathbf{q})\right)^{-1}\boldsymbol{\nu} \leq 1$

The boundary of the ellipsoid is given by the equation $\boldsymbol{\nu}^{\mathsf{T}}\left(\mathbf{J}(\mathbf{q})\mathbf{J}^{\mathsf{T}}(\mathbf{q})\right)^{-1}\boldsymbol{\nu} = 1$



$$\boldsymbol{\nu}^{\mathsf{T}}(A)^{-1}\boldsymbol{\nu} = 1$$

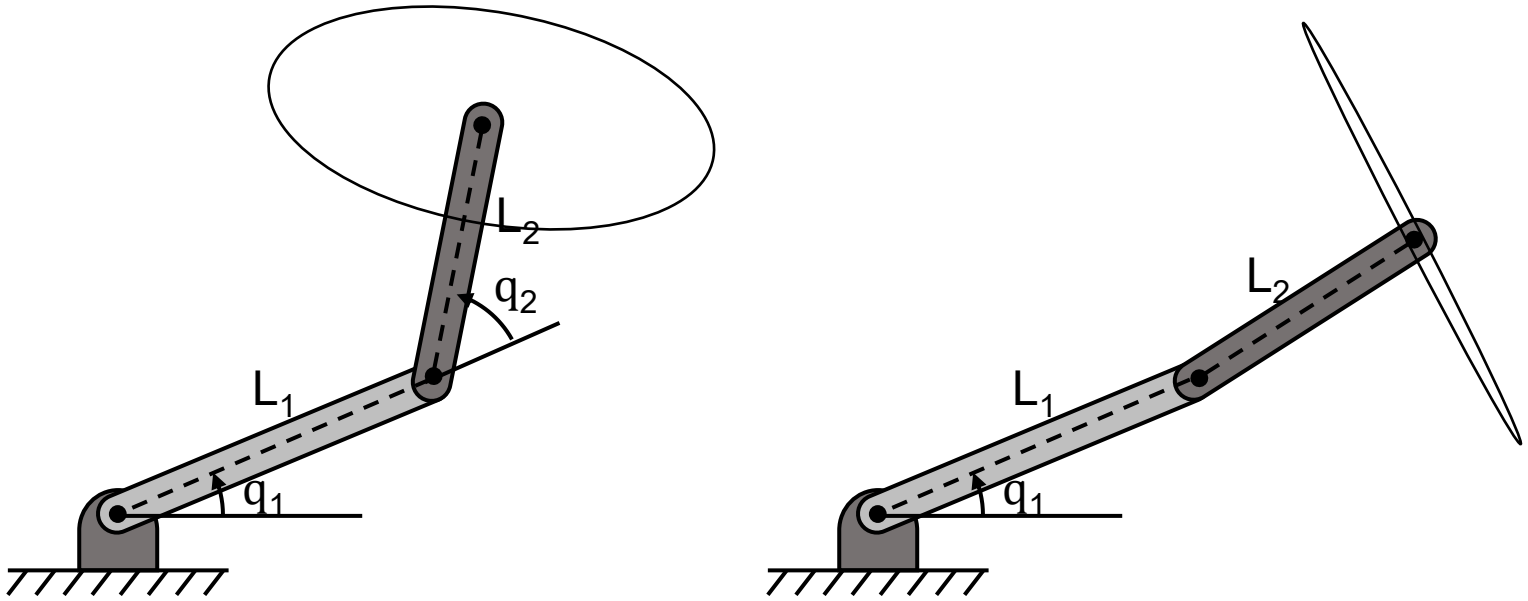
$\lambda_1, \lambda_2, \lambda_3$ are the eigen values of A

v_1, v_2, v_3 are the eigen vectors of A

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Manipulability ellipsoid at various configurations

The robot can move the fastest along the long axis direction and the slowest along the short axis direction



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Manipulability measure

- Sometimes referred to as just manipulability

- $\mathrm{m} := \frac{\sqrt{\lambda_{min}(A)}}{\sqrt{\lambda_{max}(A)}}, 0 \leq m \leq 1$ Note: some textbooks define the reciprocal as m

- When m=1, the motion is isotropic, equally possible in all directions.
- When m=0, there is a singularity. The ellipse has collapsed in one dimension.

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Example

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Relating end effector forces and joint torques

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The Jacobian can also relate forces and torques

- Let the forces and torques on the end effector be given as $\mathbf{F}_E = \begin{bmatrix} F_x \\ F_y \\ F_z \\ n_x \\ n_y \\ n_z \end{bmatrix}$

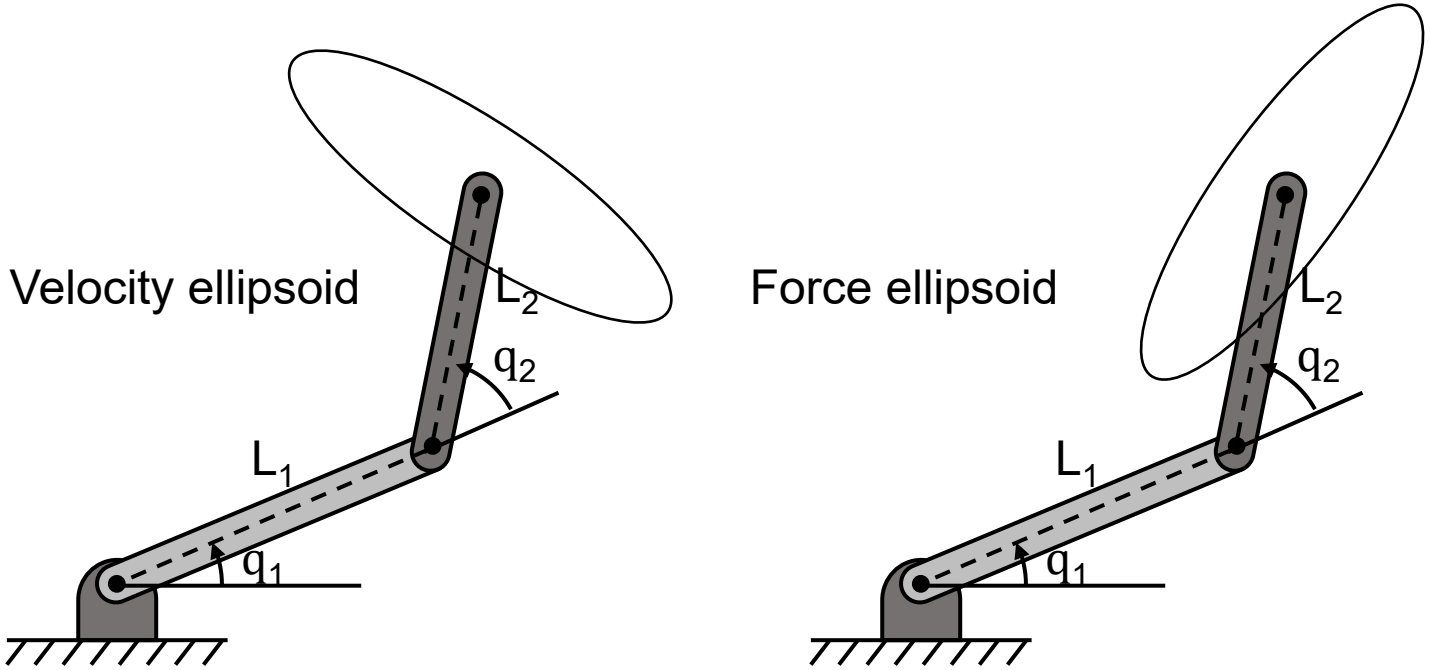
- And the torques at the joints be given by the vector $\boldsymbol{\tau}$
- The relation between the two is given as

$$\boldsymbol{\tau} = \mathbf{J}^T(\mathbf{q})\, \mathbf{F}_E$$

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One can also write down the force ellipsoid

$$\mathbf{F}^{\mathrm{T}}\left(\mathbf{J}(\mathbf{q})\mathbf{J}^{\mathrm{T}}(\mathbf{q})\right)\ \mathbf{F} \leq 1$$



Force ellipsoid and velocity ellipsoid have

- Same principal axes
- Principal axes lengths are reciprocal

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Summary

- We learnt a technique to compute the Jacobian
- We saw that the Jacobian is a useful tool
  - That relates joint velocities and spatial velocities
  - For understanding singular configurations
- We defined a measure called manipulability to indicate how close a robot's configuration is to a singularity
- Jacobian is also useful in relating forces and torques

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Next week

- Assessment 1
  - Will take place in the practical session.
  - You must be present in person to take the quiz.

- We will test your trajectory code in the prac session.

- Next lecture is on dynamics

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Assessment 1

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